

Section 8.4 Exercises

Directions: For exercises 1 - 28, graph the polar equation. Identify the name of the polar curve.

1. $r = 3\cos \theta$

2. $r = 4\sin \theta$

3. $r = 2 + 2\cos \theta$

4. $r = 2 - 2\cos \theta$

5. $r = 5 - 5\sin \theta$

6. $r = 3 + 3\sin \theta$

7. $r = 3 + 2\sin \theta$

8. $r = 7 + 4\sin \theta$

9. $r = 4 + 3\cos \theta$

10. $r = 5 + 4\cos \theta$

11. $r = 10 + 9\cos \theta$

12. $r = 1 + 3\sin \theta$

13. $r = 2 + 5\sin \theta$

14. $r = 5 + 7\sin \theta$

15. $r = 2 + 4\cos \theta$

16. $r = 5 + 6\cos \theta$

17. $r^2 = 36\cos(2\theta)$

18. $r^2 = 10\cos(2\theta)$

19. $r^2 = 4\sin(2\theta)$

20. $r^2 = 10\sin(2\theta)$

21. $r = 3\sin(2\theta)$

22. $r = 3\cos(2\theta)$

23. $r = 5\sin(3\theta)$

24. $r = 4\sin(4\theta)$

25. $r = 4\sin(5\theta)$

26. $r = -\theta$

27. $r = 2\theta$

28. $r = -3\theta$